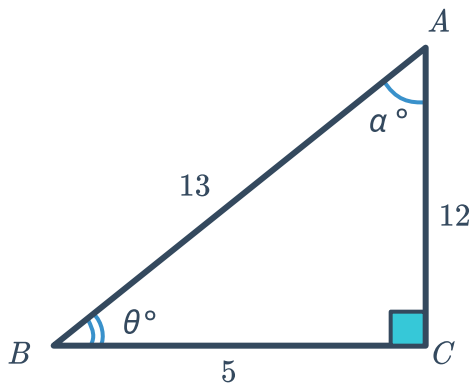


Trigonometric ratios

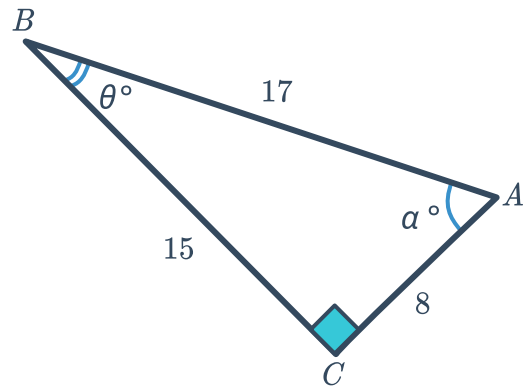


1 Evaluate $\cos \theta$ for each of the following triangles:

a

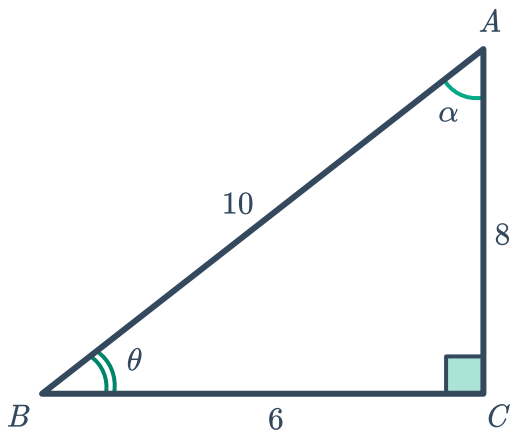


b

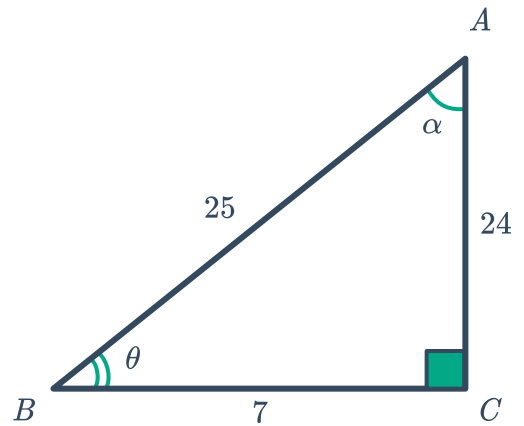


2 Evaluate $\sin \theta$ for each of the following triangles:

a



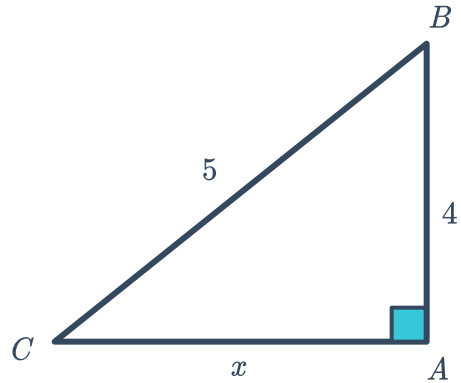
b



- 3 In the following triangle, $\sin \theta = \frac{4}{5}$:

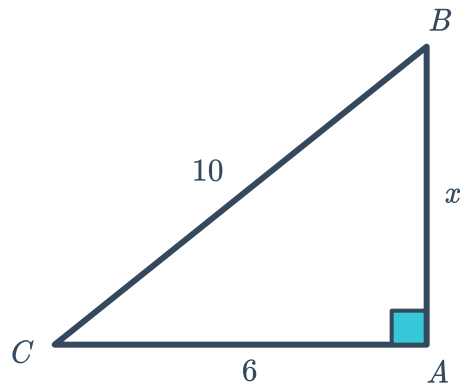


- a Which angle is represented by θ ?
- b Find $\cos \theta$.
- c Find $\tan \theta$.



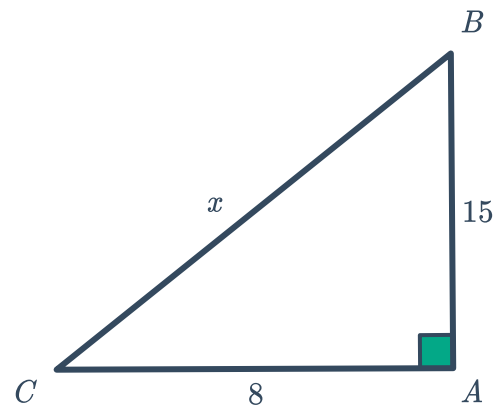
- 4 In the following triangle, $\cos \theta = \frac{6}{10}$:

- a Which angle is represented by θ ?
- b Find $\sin \theta$.
- c Find $\tan \theta$.



- 5 In the following triangle, $\tan \theta = \frac{15}{8}$:

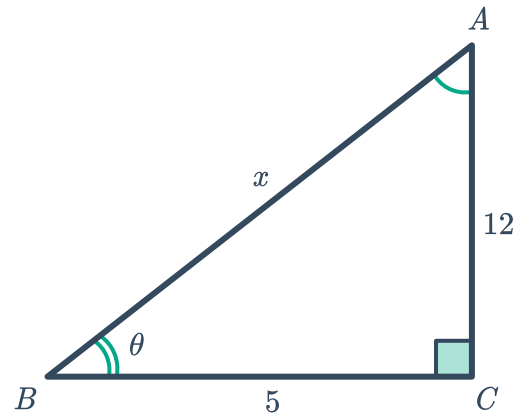
- a Which angle is represented by θ ?
- b Find $\cos \theta$.
- c Find $\sin \theta$.



6 Consider the following triangle:

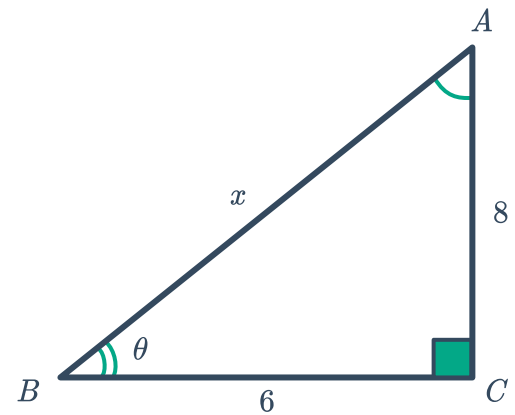


- a Find the value of x .
- b Now, find $\sin \theta$.
- c Now, find $\cos \theta$.



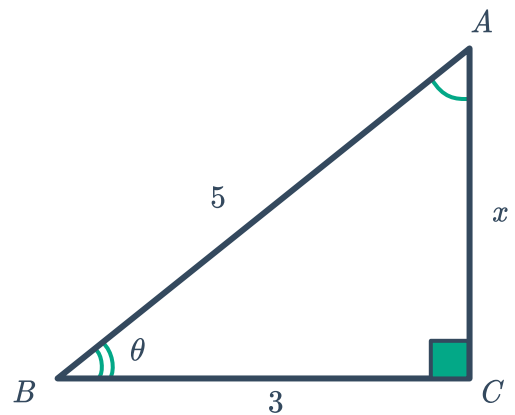
7 Consider the following triangle:

- a Find the value of x .
- b Find $\sin \theta$.
- c Find $\cos \theta$.



8 Consider the following triangle:

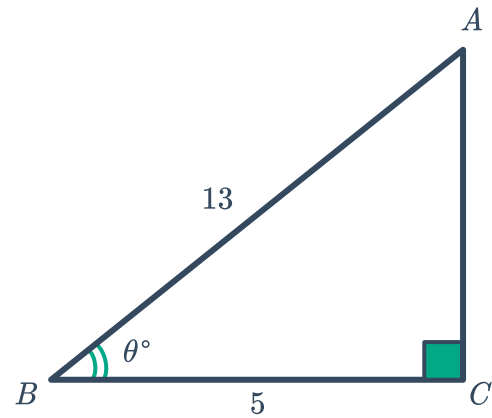
- a Find the value of x .
- b Now, find $\tan \theta$.



9 Consider the following triangle:

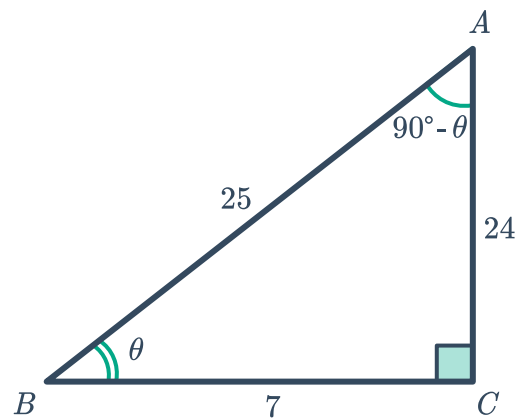


- a Find the length of AC .
- b Find $\tan \theta$.



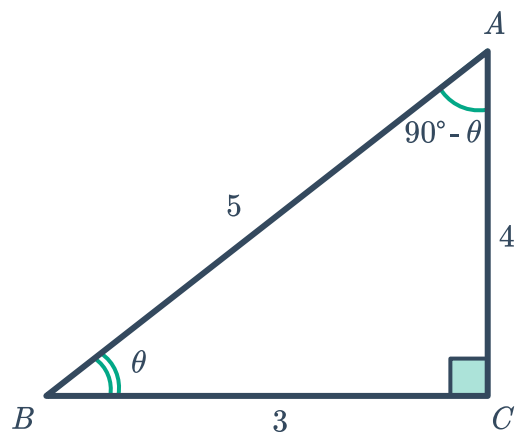
10 Consider the following triangle:

- a Find $\cos(90^\circ - \theta)$ as a fraction.
- b Find $\sin \theta$ as a fraction.
- c What do you notice?



11 Consider the following triangle:

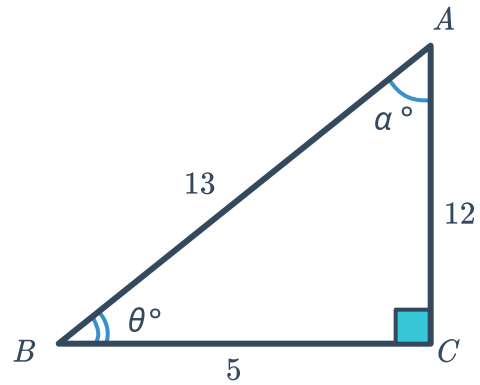
- a Find $\sin(90^\circ - \theta)$ as a fraction.
- b Find $\cos \theta$ as a fraction.
- c What do you notice?



12 Consider the following triangle:

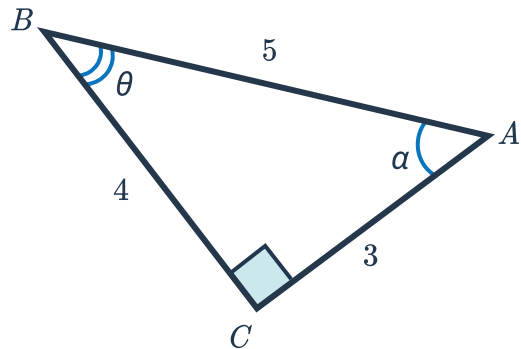


- a Evaluate $\frac{\sin \theta}{\cos \theta}$.
- b Evaluate $\tan \theta$.
- c What do you notice?

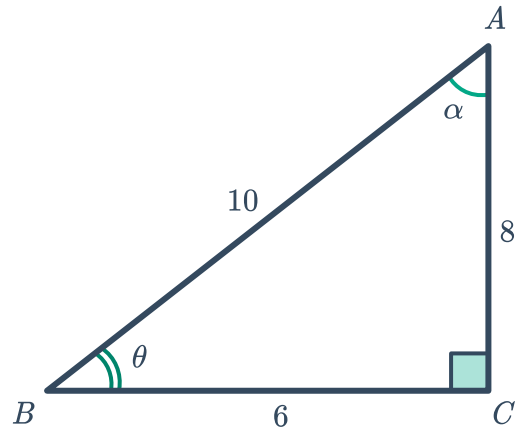


Reciprocal trigonometric ratios

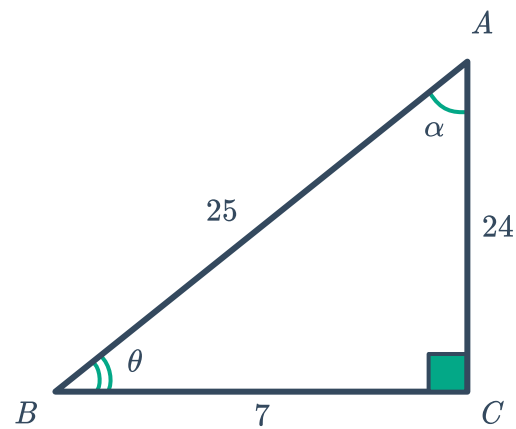
13 Find $\sec \theta$ for the following triangle:



14 Find $\csc \theta$ for the following triangle:

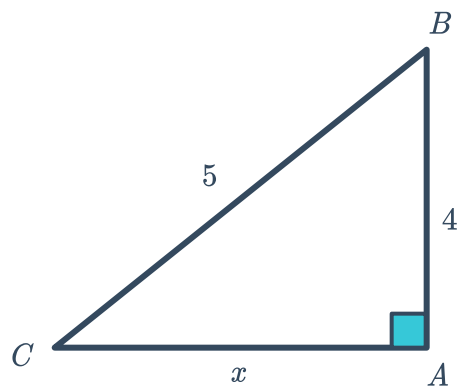


15 Find $\cot \theta$ for the following triangle:



16 In the following triangle, $\csc \theta = \frac{5}{4}$:

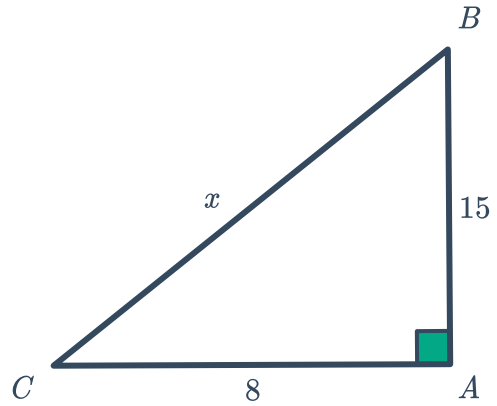
- a Which angle is represented by θ ?
- b Find the value of x .
- c Find $\sec \theta$.
- d Find $\cot \theta$.



- 17 In the following triangle, $\cot \theta = \frac{8}{15}$:

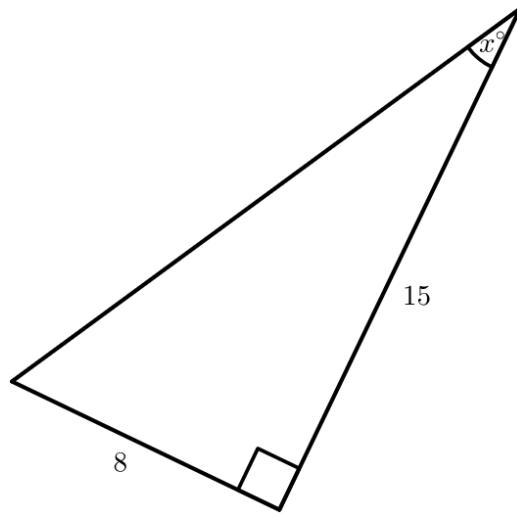


- a Which angle is represented by θ ?
- b Find the value of x .
- c Find $\sec \theta$.
- d Find $\csc \theta$.



- 18 Consider the following triangle:

- a Calculate the length of the missing side.
- b Find the following trigonometric ratios:
 - i $\sin x$
 - ii $\cos x$
 - iii $\tan x$
 - iv $\sec x$
 - v $\csc x$
 - vi $\cot x$



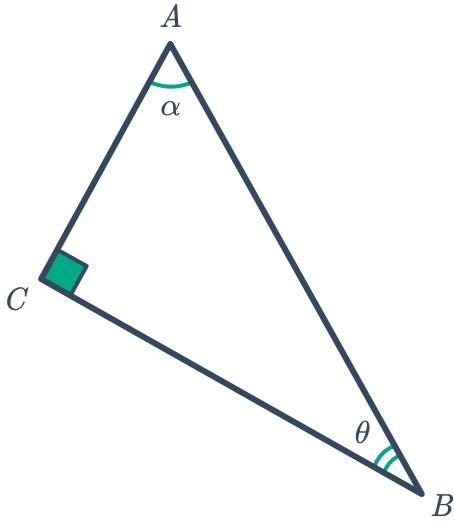
19 Find the indicated ratios for the following triangles:



a

i $\tan \theta$

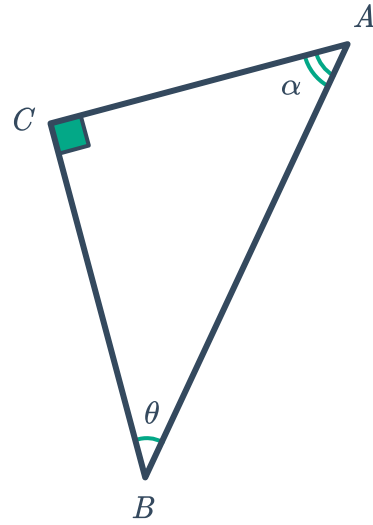
ii $\sin \alpha$



b

i $\sin \theta$

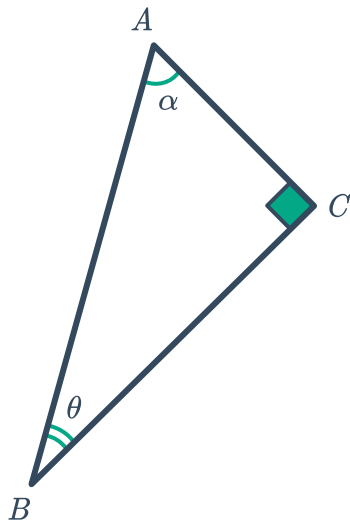
ii $\tan \alpha$



c

i $\cos \theta$

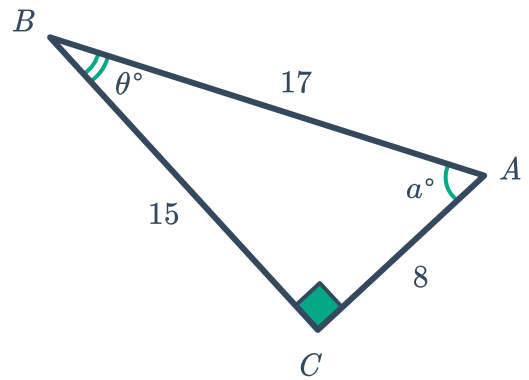
ii $\cos \alpha$



d

i $\tan \theta$

ii $\sin \alpha$



e

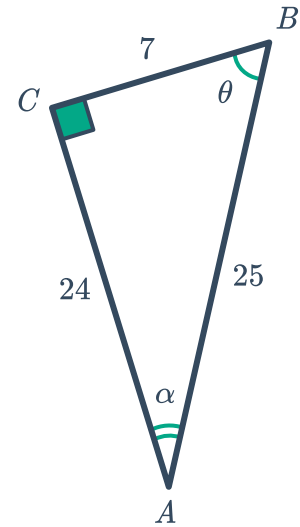
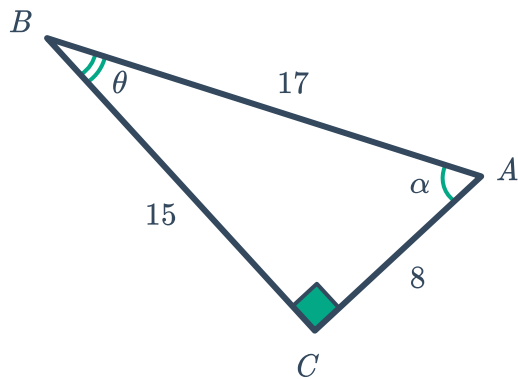
i $\cos \theta$

ii $\tan \alpha$



i $\sin \theta$

ii $\cos \alpha$



20 Consider the following triangle:

a Calculate the length of the missing side.

b Find the following trigonometric ratios:

i $\sin x$

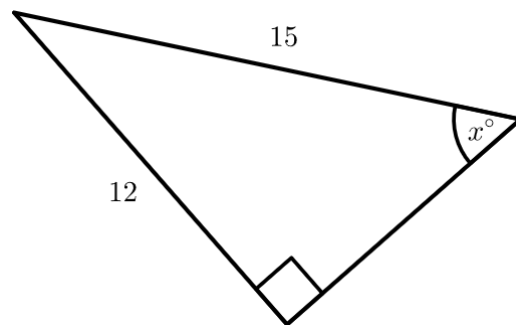
ii $\cos x$

iii $\tan x$

iv $\sec x$

v $\csc x$

vi $\cot x$



21 Consider the following triangle:



- a Given that $\tan x = \frac{3}{4}$, find $\cos x$.
- b Given that $\sin x = \frac{15}{17}$, find $\tan x$.
- c Given that $\cos x = \frac{12}{13}$, find $\sin x$.
- d Given that $\sin x = \frac{4}{5}$, find $\csc x$.
- e Given that $\tan x = \frac{7}{4}$, find $\cot x$.
- f Given that $\cos x = \frac{5}{13}$, find $\sec x$.

